

rest of the others. It's not sufficient to only check whether w is a linear combination of u, v and z .

d) Is $\{u, v, w, z\}$ linearly dependent?

Ans: No. of vectors = 4 > No. of entries in each vector = 3

$\therefore$ By Theorem 8, $\{u, v, w, z\}$ is linearly dependent $\square$

2) Suppose that $\{v_1, v_2, v_3\}$ is a linearly dependent set of vectors in $\mathbb{R}^n$ and v_4 is a vector in $\mathbb{R}^n$. Show that $\{v_1, v_2, v_3, v_4\}$ is also a linearly dependent set.

Soln: By Theorem 7, there exists some vector in $\{v_1, v_2, v_3\}$ which is a linear combination of the other vectors. WLOG, let that vector be v_3 .

$$\therefore v_3 = c_1 v_1 + c_2 v_2 \quad \text{for some } c_1, c_2 \in \mathbb{R}. \quad v_3 + 0 \cdot v_4 = (c_1 v_1 + c_2 v_2) + 0 \cdot v_4$$

$$\Rightarrow v_3 + 0 = c_1 v_1 + c_2 v_2 + 0 \cdot v_4 \Rightarrow \boxed{v_3 = c_1 v_1 + c_2 v_2 + 0 \cdot v_4} \quad \text{(Algebraic property (iii))}$$

$\therefore v_3$ is a linear combination of v_1, v_2 and v_4 . By Theorem 7, $\{v_1, v_2, v_3, v_4\}$ is a linearly dependent set. $\square$